Grade 9 Biology

Illustrated study notes · Life and Living

Cells · Body systems · Digestion · Breathing · Heart and blood · Reproduction

Contents

1. Cells: the microscope, animal and plant cells, specialised cells
2. Levels of organisation and the systems of the human body
3. The digestive system
4. The respiratory (breathing) system
5. The circulatory system: heart, blood vessels and blood
6. Human reproduction: reproductive systems, puberty, the menstrual cycle, pregnancy

These notes follow the standard Grade 9 Natural Sciences (Life and Living) content. Check them against your school's exam guideline.

1. Cells – the building blocks of life

All living things are made of **cells**. A cell is the **smallest unit of life** – the smallest thing that can carry out all the life processes (feeding, breathing, growing, getting rid of waste, reproducing). Most cells are so small that you need a **microscope** to see them.

The light microscope

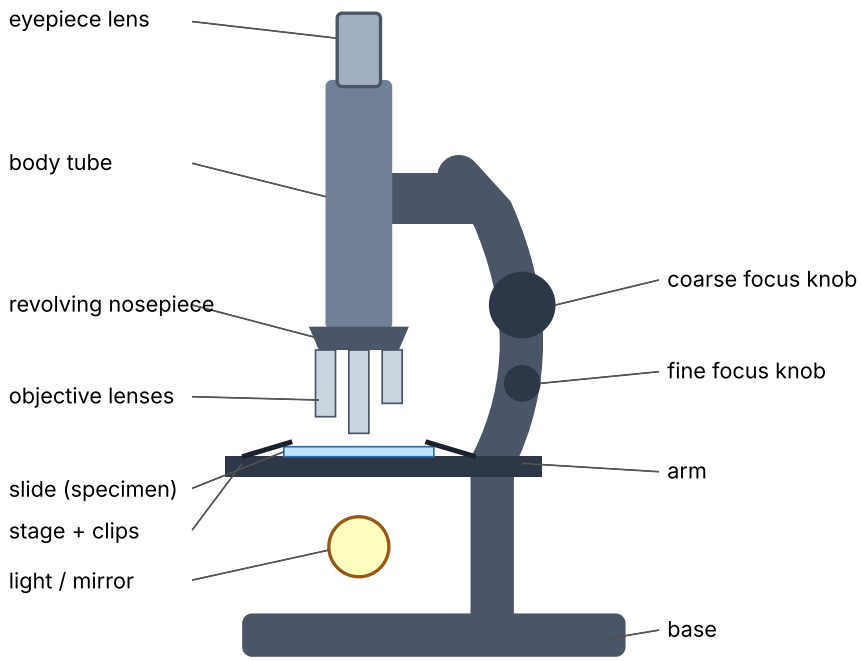


Figure 1: Parts of a light microscope

Part	Function
Eyepiece	Lens you look through; usually magnifies 10×
Objective lenses	Lenses close to the specimen; different magnifications (e.g. 4×, 10×, 40×)
Stage and clips	Flat platform that holds the slide; clips keep it still
Light / mirror	Shines light up through the specimen
Coarse focus knob	Moves the stage a lot – use first, with the lowest-power lens
Fine focus knob	Small movements to make the image sharp
Arm and base	Hold and carry the microscope (one hand on the arm, one under the base)

Total magnification = eyepiece magnification × objective magnification
Eyepiece 10× and objective 40× → 10 × 40 = **400×** (the cell looks 400 times bigger)

The animal cell

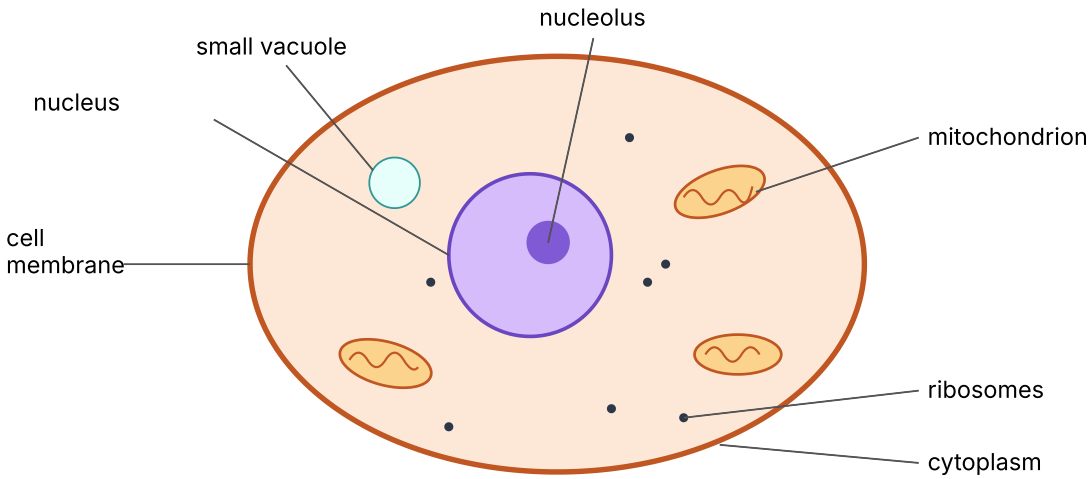


Figure 2: An animal cell (e.g. a cheek cell). Animal cells have no fixed shape.

The plant cell

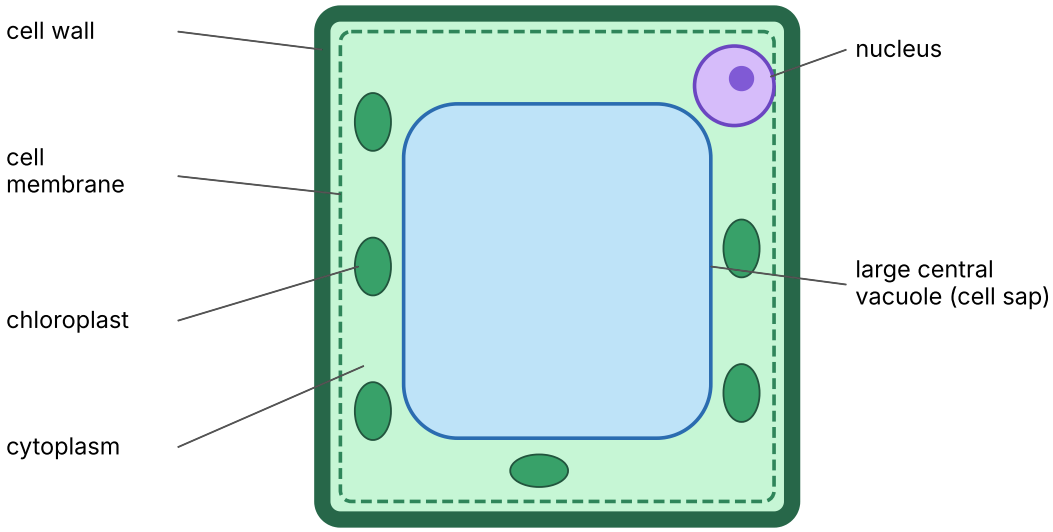


Figure 3: A plant cell (e.g. a leaf cell). The cell wall gives it a fixed, box-like shape.

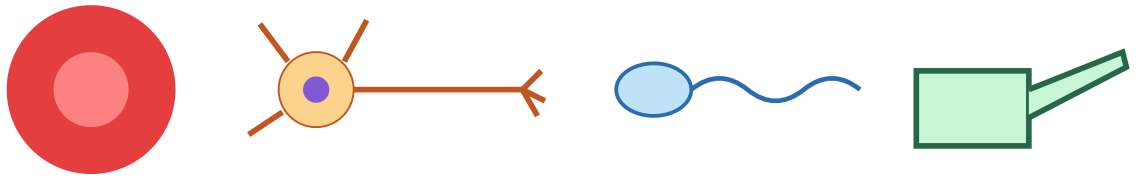
What each part does

Part	Function (job)	Animal cell	Plant cell
Cell membrane	Thin layer around the cell; controls what goes in and out	✔	✔
Cytoplasm	Jelly-like liquid where most chemical reactions happen	✔	✔
Nucleus	The "control centre"; contains the genetic material (DNA / chromosomes)	✔	✔
Mitochondria	Cellular respiration: release energy from food (glucose)	✔	✔
Ribosomes	Make proteins	✔	✔
Cell wall	Strong outer layer made of **cellulose**; gives support and shape	✘	✔
Chloroplasts	Contain green **chlorophyll**; carry out **photosynthesis** (make food)	✘	✔
Vacuole	Stores water and dissolved substances (cell sap); keeps the plant cell firm	small, many / none	one **large** central vacuole

Differences to remember: only **plant cells** have a **cell wall**, **chloroplasts** and a **large central vacuole**. Plant cells have a fixed (regular) shape; animal cells have an irregular shape.

Specialised cells

Cells in a multicellular organism are **specialised** – each type has a special **shape (structure)** that suits its **job (function)**.



Red blood cell
no nucleus, disc shape
→ carries more oxygen

Nerve cell
very long thin fibre
→ carries messages far

Sperm cell
long tail
→ swims to the egg

Root hair cell
long extension
→ absorbs more water

Figure 4: Specialised cells – the shape of each cell suits its job

2. Levels of organisation and body systems

In a multicellular organism, cells are organised into bigger and bigger groups. Each level is made of the level before it.

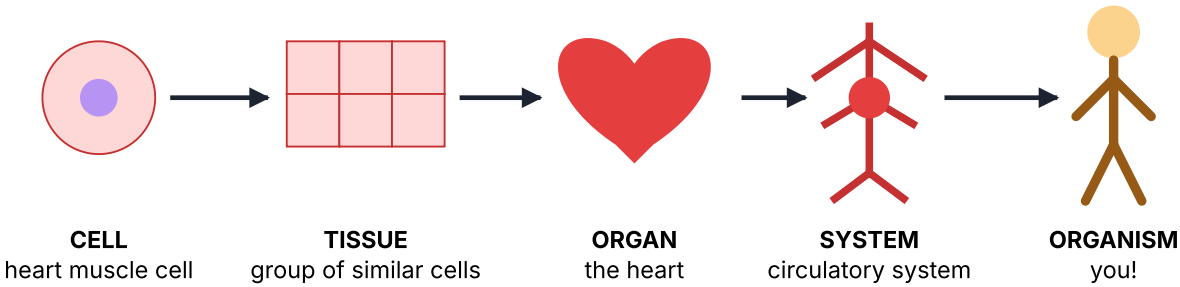


Figure 5: cell → tissue → organ → organ system → organism

- **Tissue:** a group of **similar cells** that do the same job (muscle tissue, nerve tissue, blood).
- **Organ:** different tissues working together to do a job (heart, stomach, lungs, brain).
- **Organ system:** organs that work together for a main function (digestive system).
- **Organism:** all the systems working together make a living thing.

The systems of the human body

> The systems **depend on each other**. Example: the digestive system provides glucose, the respiratory system provides oxygen, and the circulatory system carries both to every cell so the cells can release energy.

System	Main organs	Main job
Skeletal	Bones, joints, cartilage	Support, protection (skull protects the brain), movement, makes blood cells
Muscular	Muscles, tendons	Movement (muscles pull on bones)
Digestive	Mouth, stomach, intestines, liver	Breaks food down so it can be absorbed
Respiratory	Nose, trachea, lungs	Takes in oxygen, removes carbon dioxide (gas exchange)
Circulatory	Heart, blood vessels, blood	Transports oxygen, food, waste and heat around the body
Excretory	Kidneys, bladder, skin, lungs	Removes waste (urine, sweat, CO_2)
Nervous	Brain, spinal cord, nerves	Controls the body; carries messages; reacts to stimuli
Endocrine	Glands (pituitary, thyroid, pancreas, ovaries, testes)	Makes **hormones** – chemical messengers in the blood
Reproductive	Testes / ovaries, uterus	Produces offspring (babies)
Immune	White blood cells, lymph nodes	Fights disease

3. The digestive system

Digestion is the process of **breaking down large food molecules into small, soluble molecules** that can be absorbed into the blood. Food travels through a long tube called the **alimentary canal** (gut), from the mouth to the anus.

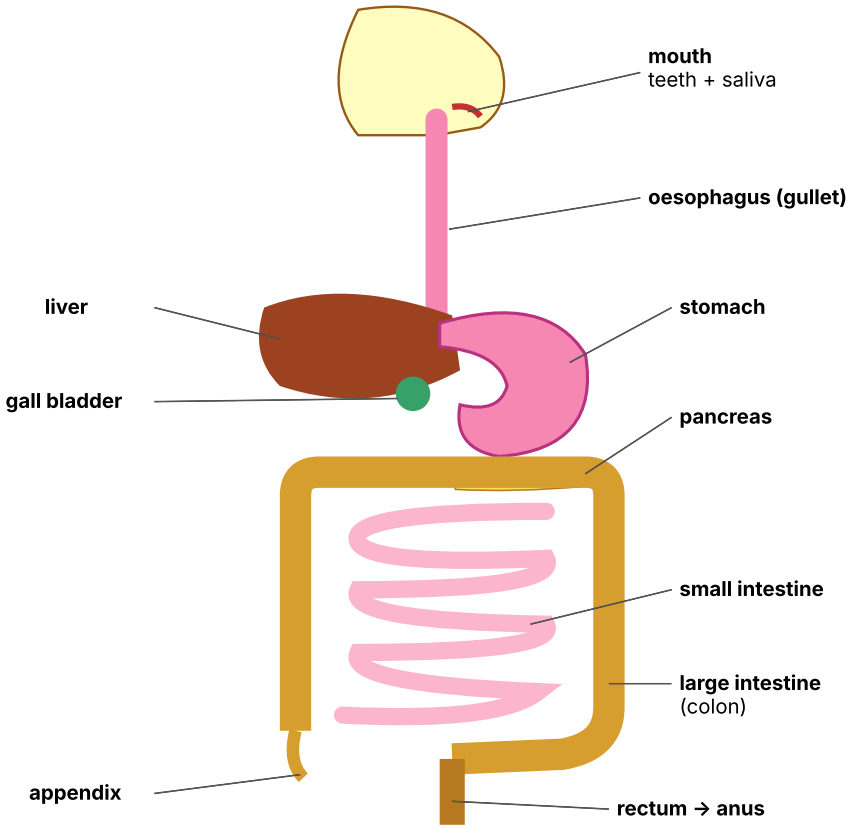


Figure 6: The human digestive system

Two types of digestion

- **Mechanical (physical) digestion:** breaking food into **smaller pieces** – chewing with teeth, churning in the stomach. This gives a bigger surface area for enzymes to work on.
- **Chemical digestion: enzymes** break big food molecules into small molecules. Enzymes are proteins that speed up chemical reactions.

The journey of food

Organ	What happens
Mouth	**Ingestion** (taking in food). Teeth chew (mechanical). Saliva moistens food and contains **amylase**, which starts digesting starch. The tongue rolls food into a ball (**bolus**).
Oesophagus	Muscle tube that pushes food down to the stomach by **peristalsis** – waves of muscle squeezing.
Stomach	Muscular bag that **churns** food. Gastric juice contains **hydrochloric acid** (kills germs) and **pepsin** (starts digesting protein). Food becomes a thick liquid called **chyme**.
Liver and gall bladder	Liver makes **bile**, stored in the gall bladder. Bile breaks fat into small droplets (**emulsification**).
Pancreas	Makes pancreatic juice with enzymes that digest starch, protein and fat. Also makes the hormone **insulin**.
Small intestine	Digestion is completed. **Absorption**: small food molecules pass through the wall into the blood. About 6 m long, lined with millions of **villi**.
Large intestine	**Water** (and some minerals) is absorbed back into the body. Undigested food becomes faeces.
Rectum and anus	Faeces are stored in the rectum and passed out through the anus (**egestion**).

Absorption in the small intestine: villi

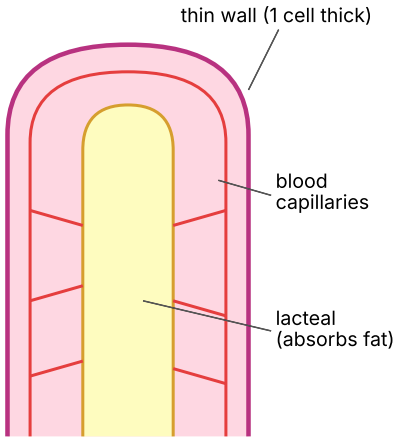


Figure 7: A villus

The inside wall of the small intestine is covered with millions of tiny finger-like folds called **villi** (one is a **villus**). They are perfect for absorption because they:

- give a **very large surface area**,
- have **very thin walls** (one cell thick), so food passes through quickly,
- have a **good blood supply** (capillaries) to carry food away.

Glucose and amino acids go into the **blood**; fats go into the **lacteal**.

Food types and what they are broken into

Food type	Found in	Digested into	Used for
Carbohydrates (starch)	Bread, rice, maize meal, potatoes	**Glucose**	Energy
Proteins	Meat, fish, eggs, beans	**Amino acids**	Growth and repair
Fats (lipids)	Butter, oil, nuts	**Fatty acids + glycerol**	Stored energy, insulation

> **Five stages of digestion:** ingestion (eating) → digestion (breaking down) → absorption (into blood) → assimilation (cells use the food) → egestion (removing undigested waste).

Problems: heartburn (acid moves up the oesophagus), ulcers, constipation (too little fibre and water), diarrhoea (often from germs in dirty water – causes dehydration), obesity and malnutrition. A **balanced diet** with fibre, fruit, vegetables and water keeps the system healthy.

4. The respiratory (breathing) system

The respiratory system brings **oxygen** into the body and removes **carbon dioxide**. Every cell needs oxygen to release energy from glucose (**cellular respiration**).

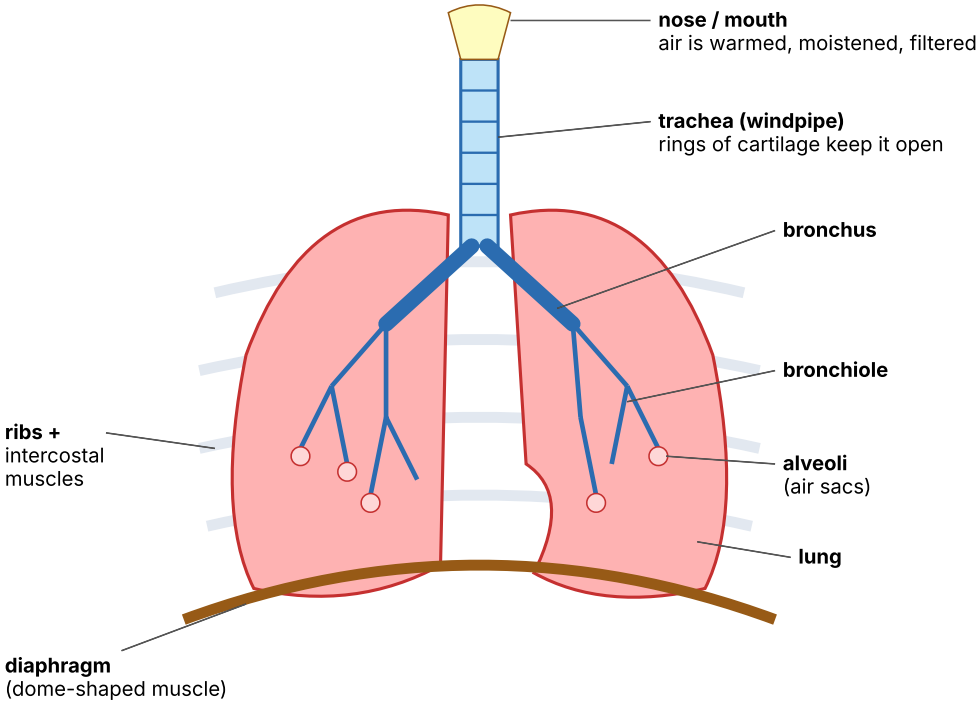


Figure 8: The human respiratory system

The path of air: nose / mouth → **pharynx** → **larynx** (voice box) → **trachea** → **bronchi** (one to each lung) → **bronchioles** → **alveoli**. The **epiglottis** is a flap that closes the trachea when you swallow, so food doesn't go into the lungs.

Gas exchange in the alveoli

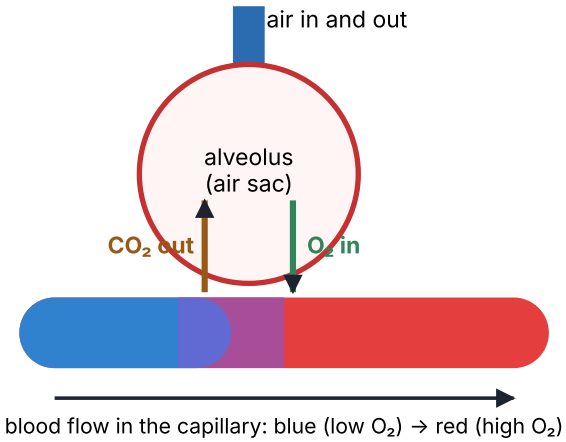


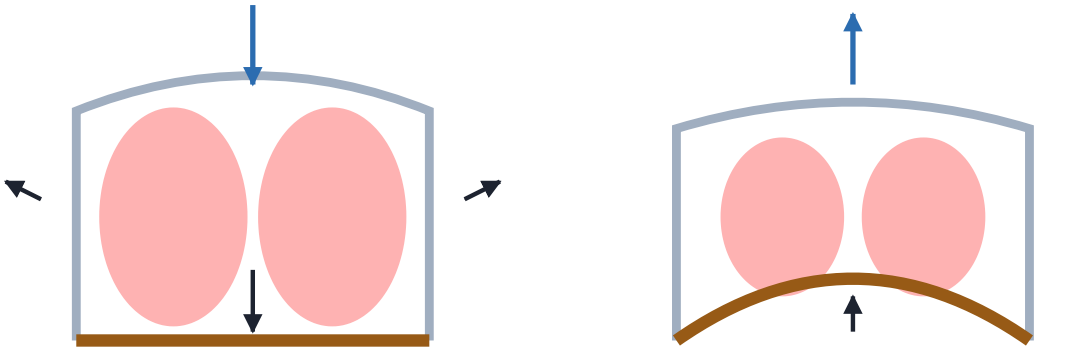
Figure 9: O_2 moves from the air into the blood; CO_2 moves from the blood into the air

Gases move by **diffusion** – from where there is a **lot** of a substance to where there is **less**.

- **Oxygen** diffuses from the air in the alveolus **into the blood**.
- **Carbon dioxide** diffuses from the blood **into the alveolus** and is breathed out.

Alveoli are good at gas exchange because they have: a **huge surface area** (about 300 million alveoli), **very thin walls** (one cell thick), a **moist lining**, and a **rich blood supply**.

Breathing in and out



BREATHING IN (inhalation)
diaphragm flattens, ribs move up and out

BREATHING OUT (exhalation)
diaphragm domes up, ribs move down and in

Figure 10: How the diaphragm and ribs change the size of the chest

	Breathing in (inhalation)	**Breathing out (exhalation)**
Diaphragm	Contracts and **flattens** (moves down)	Relaxes and **domes up**
Intercostal muscles / ribs	Contract; ribs move **up and out**	Relax; ribs move **down and in**
Volume of chest	**Increases**	**Decreases**
Pressure inside the lungs	**Decreases** → air rushes **in**	**Increases** → air is pushed **out**

Gas	**Inhaled air**	**Exhaled air**
Oxygen	about 21%	about 16%
Carbon dioxide	about 0.04%	about 4%
Water vapour	less	more (that's why breath mists a mirror)

> **Breathing ≠ respiration.** **Breathing** is moving air in and out of the lungs. **Cellular respiration** happens in the mitochondria of every cell: **glucose + oxygen → carbon dioxide + water + energy**.

Diseases: asthma (airways narrow, hard to breathe), **tuberculosis (TB)** (bacteria infect the lungs; spreads through coughing), **pneumonia**, **bronchitis**, **lung cancer** and **emphysema** (alveoli are damaged – mainly caused by smoking).

5. The circulatory system

The circulatory system is the body's **transport system**. The **heart** pumps **blood** through **blood vessels** to carry oxygen, food (glucose), water, hormones and heat to every cell, and to take carbon dioxide and other waste away.

The heart

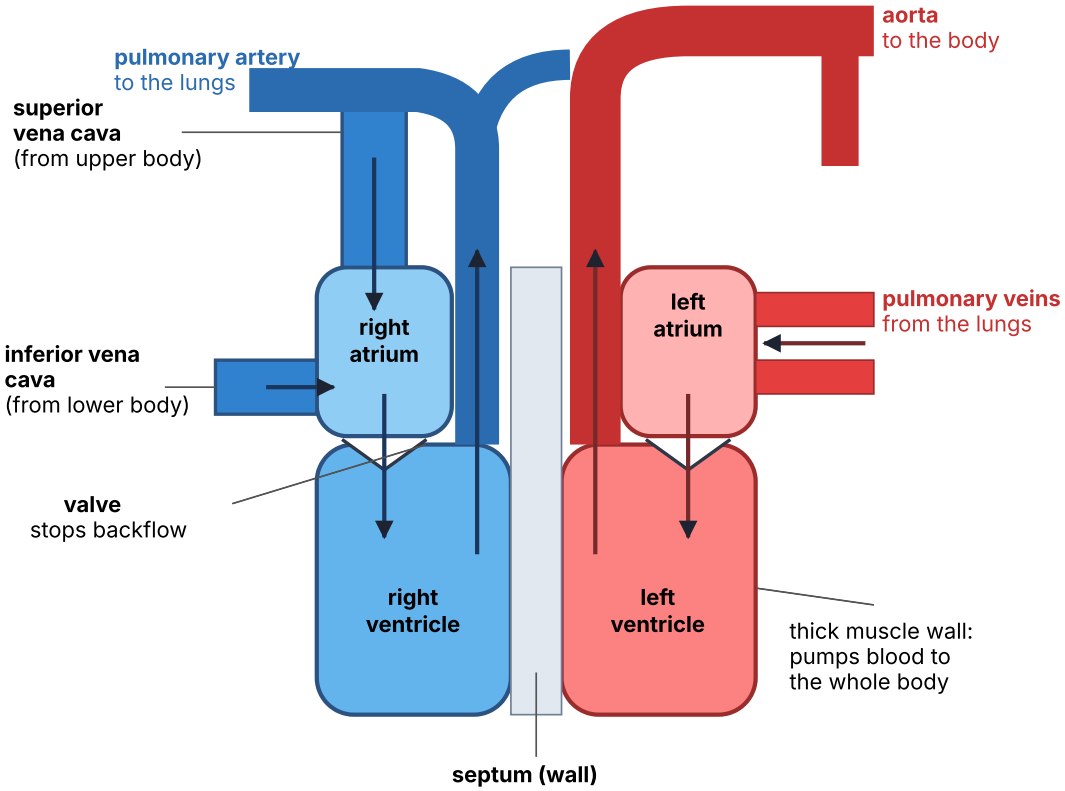


Figure 11: The heart, seen from the front. The heart's left side is on the right of the drawing, as if you are facing the person.

- The heart has **4 chambers**: two **atria** at the top (receive blood) and two **ventricles** at the bottom (pump blood out).
- The **right side** pumps **deoxygenated** blood to the **lungs**. The **left side** pumps **oxygenated** blood to the **whole body**.
- The **left ventricle** has the **thickest wall** because it must pump blood all around the body.
- The **septum** separates the two sides so oxygenated and deoxygenated blood don't mix.
- **Valves** make sure blood flows in **one direction only** (the "lub-dub" sound is valves closing).
- The heart is made of **cardiac muscle**, which never gets tired. Its own blood supply comes from the **coronary arteries**.

Double circulation: the path of blood

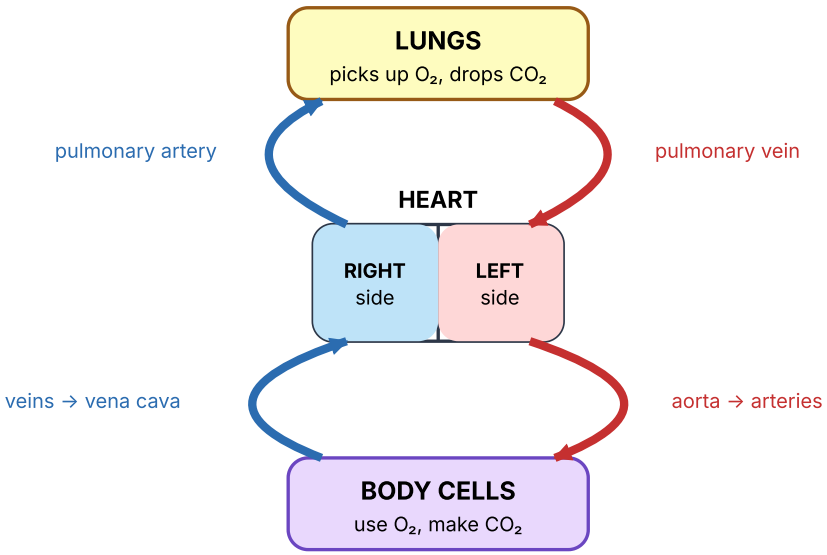


*Figure 12: Blood passes through the heart **twice** on every trip around the body*

> **Learn this order:** body → vena cava → **right atrium** → **right ventricle** → pulmonary artery → **lungs** → pulmonary vein → **left atrium** → **left ventricle** → aorta → body.

- **Pulmonary circulation:** heart → lungs → heart (to pick up oxygen).
- **Systemic circulation:** heart → body → heart (to deliver oxygen and food).

Blood vessels

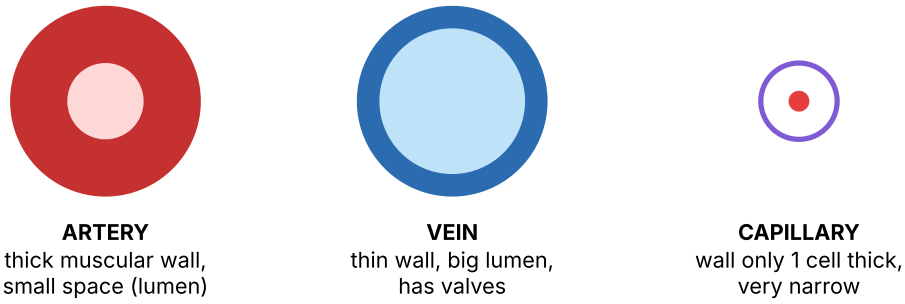


Figure 13: Cross-sections of the three types of blood vessels

	Arteries	Veins	Capillaries
Direction	**Away** from the heart (A for Away)	**Towards** the heart	Connect arteries to veins, between the cells
Blood	Oxygenated (except pulmonary artery)	Deoxygenated (except pulmonary vein)	Gives O_2 and food to cells, takes waste
Pressure	High – the pulse you feel	Low	Low
Wall	Thick, muscular, elastic	Thinner; valves stop backflow	One cell thick – substances pass through

What blood is made of

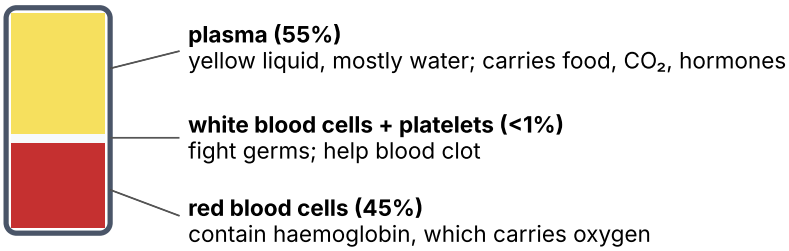


Figure 14: Blood separated in a test tube

Part	Job
Red blood cells	Carry **oxygen** using **haemoglobin** (contains iron). No nucleus, disc-shaped.
White blood cells	**Fight infection**: engulf germs or make antibodies.
Platelets	Help blood **clot** to stop bleeding.
Plasma	Liquid that carries dissolved food, CO_2, waste, hormones and heat.

Diseases and health: high blood pressure (hypertension), **heart attack** (a coronary artery is blocked by fatty deposits), **stroke** (blood supply to the brain is blocked), **anaemia** (too few red blood cells or too little iron – tired and pale). Keep your heart healthy: exercise, eat less fat, salt and sugar, don't smoke, keep a healthy weight.

6. Human reproduction

Reproduction is how living things produce offspring. Humans reproduce **sexually**: a male **sex cell (sperm)** joins a female **sex cell (egg / ovum)**. Sex cells are made in the **gonads** – the **testes** in males and the **ovaries** in females.

Puberty

Puberty is the stage (usually between 9 and 16 years) when the body becomes able to reproduce. It is controlled by **hormones**: **testosterone** in boys and **oestrogen** and **progesterone** in girls.

Changes in boys	Changes in girls	Both
Voice deepens Facial and chest hair Shoulders broaden, muscles develop Testes start making sperm Penis and testes grow	Breasts develop Hips widen Ovaries start releasing eggs Menstruation (periods) begins	Growth spurt Pubic and underarm hair Sweat more, body odour Pimples (acne) Emotional changes, mood swings

The male reproductive system

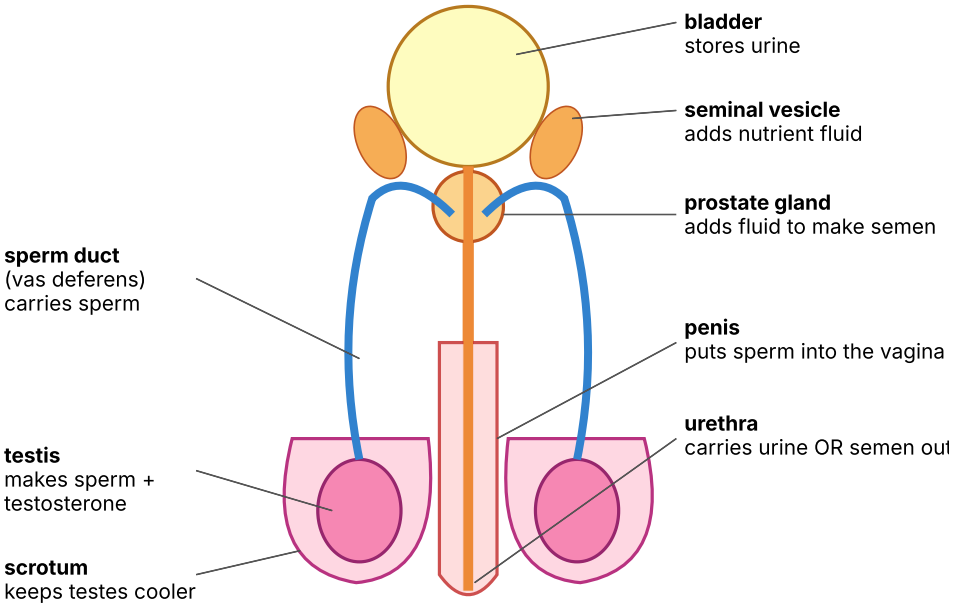


Figure 15: The male reproductive system (front view, simplified)

Path of sperm: testis → sperm duct → (seminal vesicle and prostate add fluid → **semen**) → urethra → out through the penis. Testes hang outside the body in the scrotum because sperm develop best slightly **below body temperature**.

The female reproductive system

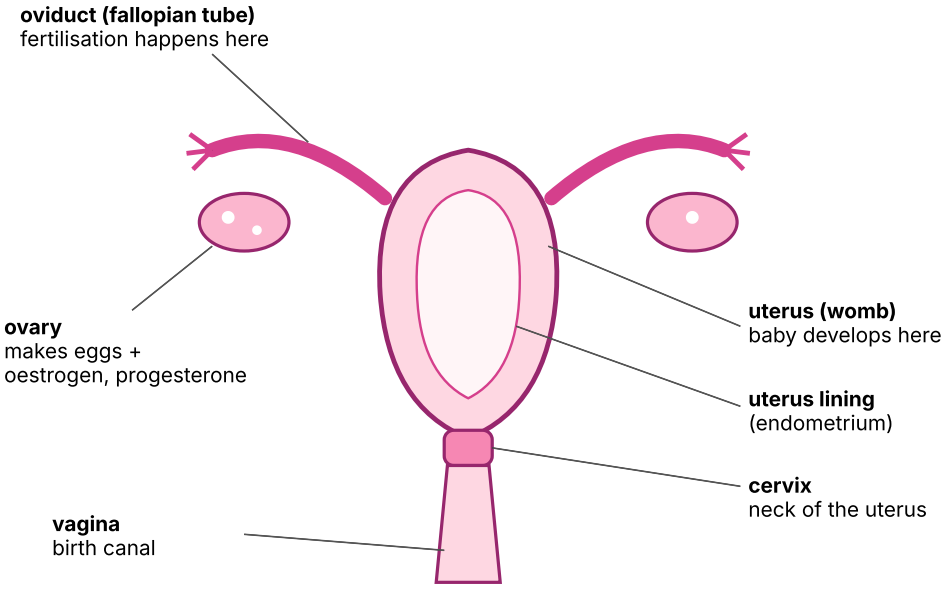


Figure 16: The female reproductive system (front view, simplified)

The menstrual cycle

About every **28 days** the female body prepares for a possible pregnancy. This is the **menstrual cycle**.

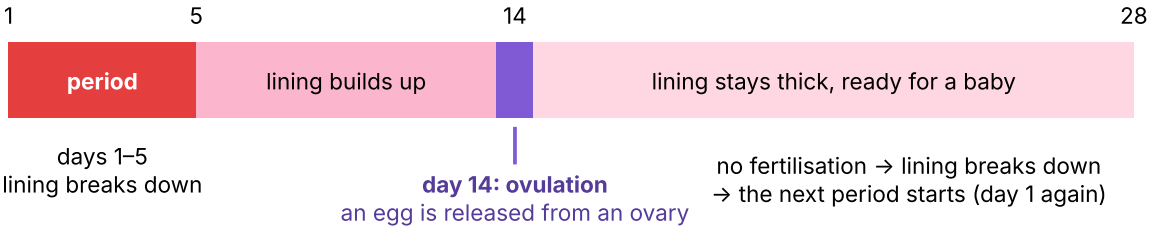
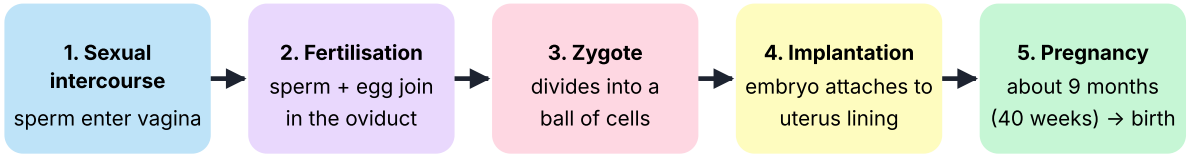


Figure 17: The 28-day menstrual cycle

1. **Days 1–5 – menstruation (period):** the thick lining of the uterus breaks down and passes out through the vagina with some blood.
2. **Days 6–13:** the uterus lining **thickens** again (controlled by oestrogen). An egg ripens in an ovary.
3. **About day 14 – ovulation:** a mature egg is released from an ovary into the oviduct. This is when a woman is most likely to fall pregnant.
4. **Days 15–28:** the lining stays thick (progesterone). If the egg is **not fertilised**, it dies, the lining breaks down and a new cycle starts. If it **is** fertilised, there is no period – the woman is pregnant.

Fertilisation, pregnancy and birth



embryo (first 8 weeks) → foetus (from about 9 weeks until birth)

Figure 18: From fertilisation to birth

- **Fertilisation:** one sperm joins the egg in the **oviduct**. The fertilised egg is a **zygote**. It contains half its genetic material from the father and half from the mother.
- The zygote divides many times as it moves to the uterus and **implants** in the thick lining.
- The **placenta** forms: it lets food and oxygen pass from the mother's blood to the baby, and waste from the baby to the mother. The baby is joined to the placenta by the **umbilical cord**. The **amniotic fluid** protects the baby from bumps.
- Alcohol, smoking and drugs pass through the placenta and can harm the baby (e.g. foetal alcohol syndrome).
- **Birth:** the uterus muscles contract, the cervix opens and the baby is pushed out through the vagina.

Preventing pregnancy and STIs

Method	How it works
Abstinence	Not having sex – the only 100% effective method against pregnancy and STIs
Condom (male / female)	Barrier that stops sperm reaching the egg; also protects against STIs and HIV
Contraceptive pill / injection / implant	Hormones stop ovulation; do **not** protect against STIs
IUD (loop)	Small device in the uterus that stops implantation

Sexually transmitted infections (STIs) spread through sexual contact: HIV, syphilis, gonorrhoea, chlamydia, herpes. Many have no symptoms at first. Prevent them with abstinence, condoms and testing; treat them early at a clinic.

> **Teenage pregnancy** can lead to dropping out of school, health risks for mother and baby, and money problems. Delay sex, get correct information, and talk to a trusted adult or clinic.

Grade 9 Physics

Illustrated study notes · Electricity · Forces · Magnetism · Electrostatics
November 2026 exam (1½ hours, 75 marks)

Contents

1. Formula sheet and how to do calculations
2. Electricity: cells, energy, generating electricity, the national grid, cost
3. Electricity: circuit components, current, voltage, resistance
4. Electricity: series and parallel circuits
5. Electricity: what affects resistance
6. Forces: types of forces, force diagrams, mass and weight
7. Magnetism: poles, magnetic fields, magnetic materials, Earth's field
8. Electrostatics: charges, charging by rubbing, conservation of charge

Also learn the definitions on your class definition page. These notes use **g = 10 N/kg**; use 9.8 N/kg if your teacher does.

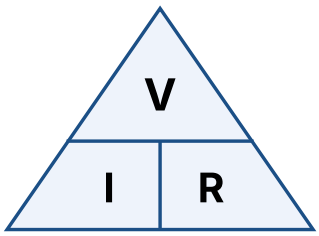
1. Formula sheet

Every calculation in this paper uses one of these formulas. Learn the **symbol** and the **unit** of each quantity.

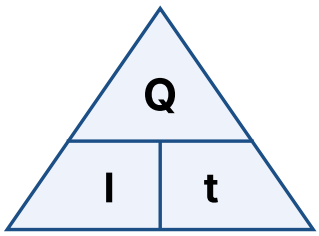
Quantity	Symbol	Unit	Formula	Rearranged
Charge	Q	coulomb (C)	**Q = I × t**	I = Q ÷ t t = Q ÷ I
Potential difference (voltage)	V	volt (V)	**V = I × R**	I = V ÷ R R = V ÷ I
Energy delivered by a cell	W	joule (J)	**W = V × Q**	V = W ÷ Q Q = W ÷ V
Weight	w	newton (N)	**w = m × g**	m = w ÷ g
Electrical energy used	E	kilowatt-hour (kWh)	**E = P (kW) × t (h)**	
Cost of electricity	–	rand (R)	**cost = kWh × price per kWh**	

The triangle trick

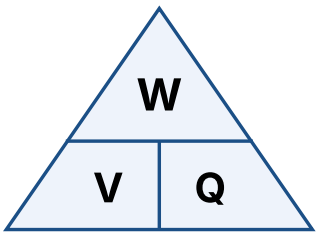
All these formulas have the form **A = B × C**. Put A on top of a triangle and B and C at the bottom. Cover the one you want: if the other two are **side by side**, multiply them; if one is **above** the other, divide.



V = I × R · I = V ÷ R · R = V ÷ I



Q = I × t · I = Q ÷ t · t = Q ÷ I



W = V × Q · V = W ÷ Q · Q = W ÷ V

Unit conversions you must do first

Change	How	Example
minutes → seconds	× 60	3 min = 180 s
hours → seconds	× 3 600	2 h = 7 200 s
milliampere (mA) → ampere (A)	÷ 1 000	250 mA = 0.25 A
gram (g) → kilogram (kg)	÷ 1 000	500 g = 0.5 kg
watt (W) → kilowatt (kW)	÷ 1 000	2 000 W = 2 kW

How to get full marks for a calculation: (1) write the **formula**, (2) **substitute** the numbers in the correct units, (3) write the **answer with its unit**. A missing unit loses a mark.

Example: A current of 0.5 A flows for 2 minutes. Calculate the charge.

t = 2 × 60 = 120 s

Q = I × t

Q = 0.5 × 120

Q = 60 C

2. Cells, energy and generating electricity

What is a cell?

A **cell** is a small store of **chemical potential energy**. When it is connected in a **closed circuit**, chemicals inside it react and push charges (electrons) around the circuit. In this way the cell changes **chemical energy → electrical energy**.

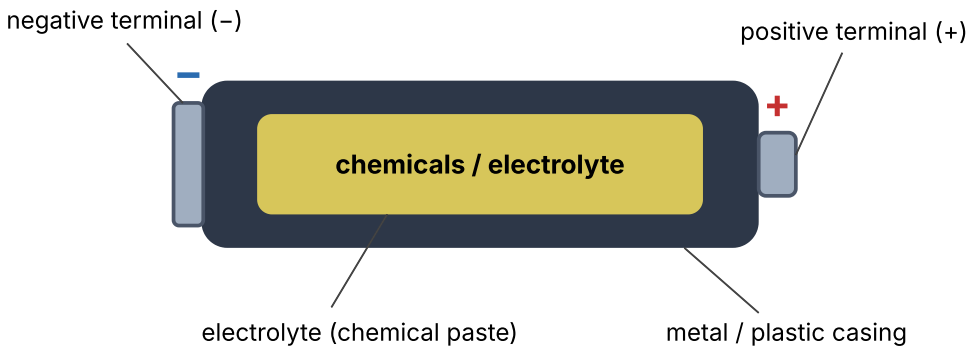


Figure 1: Inside a dry cell (like an AA torch battery, 1.5 V)

- The chemicals only react when the circuit is **closed** – that is why a cell lasts longer when the switch is off.
- When all the chemicals have reacted, the cell is **flat** and can't push charge any more.
- The **voltage** of a cell tells you how much energy it gives every coulomb of charge: a 1.5 V cell gives **1.5 J to every 1 C**.

Cells vs batteries

A **battery** is **two or more cells connected together**. When cells are joined in series (+ to −), their voltages **add up**.

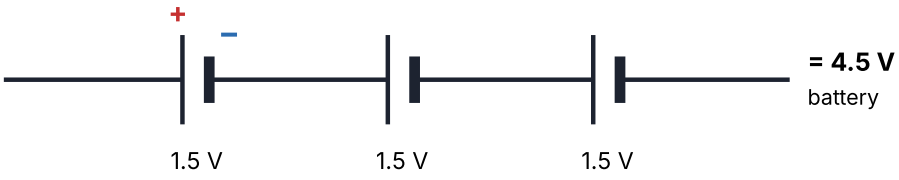


Figure 2: Three 1.5 V cells in series make a 4.5 V battery. Long thin line = positive (+); short thick line = negative (−).

Energy changes in a circuit

Energy is **never created or destroyed** – it only changes from one form to another. In a torch: chemical (cell) → electrical (wires) → light + heat (bulb).

Device	Energy change
Cell / battery	chemical → electrical
Light bulb	electrical → light + heat
Kettle, heater, iron, toaster	electrical → heat
Fan, electric motor, drill	electrical → kinetic (movement)
Buzzer, speaker, radio	electrical → sound
Charging a phone	electrical → chemical (stored in the battery)
Solar panel	light → electrical

Energy delivered by a cell: W = V × Q
A 1.5 V cell gives a current of 0.2 A for 5 minutes. How much energy does it deliver?
Step 1 – charge: Q = I × t = 0.2 × 300 = 60 C
Step 2 – energy: W = V × Q = 1.5 × 60 = **90 J**

How a power station makes electricity

Almost all power stations work the same way: something turns a **turbine**, and the turbine turns a **generator**. Inside the generator, **magnets spin near coils of copper wire**, and this movement produces an electric current.

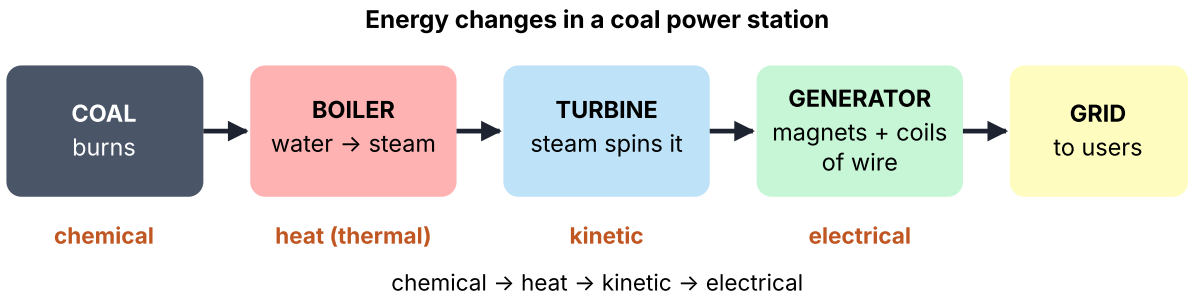


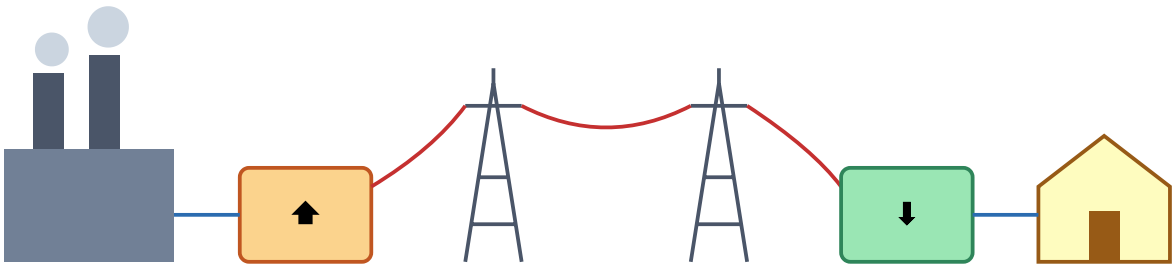
Figure 3: A coal-fired power station. Most of South Africa's electricity is made this way (Eskom).

Methods of generating electricity

Method	What turns the turbine?	Energy change	Renewable?	Good / bad
Coal	Steam from burning coal	chemical → heat → kinetic → electrical	No	+ SA has lots of coal, works day and night – air pollution, CO_2 (global warming), uses lots of water
Nuclear (Koeberg, Cape Town)	Steam heated by splitting uranium atoms	nuclear → heat → kinetic → electrical	No	+ no CO_2, lots of energy – dangerous radioactive waste, expensive
Hydro-electric	Falling water from a dam	gravitational potential → kinetic → electrical	Yes	+ clean – needs big rivers; SA is dry
Wind	Wind turns the blades	kinetic → electrical	Yes	+ clean – only when the wind blows
Solar	No turbine – solar panels change light directly	light → electrical	Yes	+ clean, lots of sun in SA – only in daylight, panels are costly

The national grid

The **national grid** is the huge network of power lines, transformers and substations that carries electricity from power stations to homes, schools and factories all over South Africa.



High voltage → small current → less energy lost as heat in the long wires

Figure 4: How electricity gets from the power station to your home

Why such a high voltage? At a high voltage, the same energy can be sent with a much **smaller current**. A smaller current heats the wires less, so **less energy is wasted** over hundreds of kilometres. The voltage is then lowered (stepped down) to a safe 230 V before it reaches homes.

Load shedding happens when people need more electricity than the power stations can make. Eskom switches areas off in turns so that the whole grid does not collapse.

Cost of electricity

- **Power** is how fast an appliance uses energy, in watts (W). 1 kilowatt (kW) = 1 000 W.
- Electricity is sold in **kilowatt-hours (kWh)**, also called **units**. 1 kWh = using 1 kW for 1 hour.
- **Energy (kWh) = power (kW) × time (hours)** and **Cost = kWh × price per kWh**

Example: A 2 000 W heater is on for 3 hours. Electricity costs R2.50 per kWh.
Power = 2 000 ÷ 1 000 = 2 kW
Energy = 2 kW × 3 h = 6 kWh
Cost = 6 × R2.50 = **R15.00**

Saving electricity: appliances that make **heat** (geyser, kettle, heater, stove) use the most. Switch off lights and appliances, use LED bulbs, put a blanket on the geyser, use a solar water heater, and boil only the water you need.

3. Circuits: components and key terms

An **electric circuit** is a **closed path** that charge can flow around. It needs an energy source (cell), connecting wires, and something that uses the energy (bulb, resistor). If there is a gap anywhere – an open switch or a broken wire – it is an **open circuit** and no current flows.

Circuit symbols

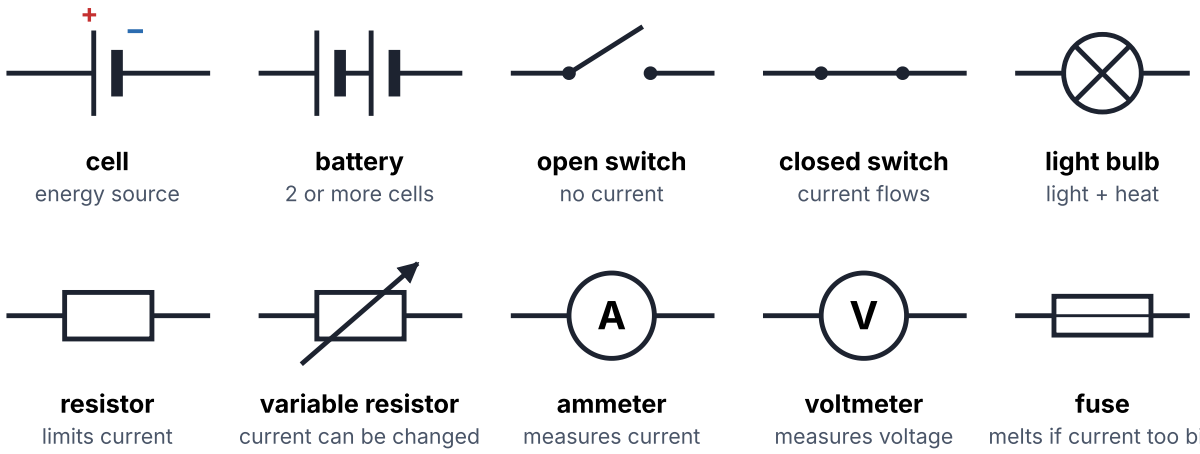


Figure 5: Circuit symbols you must know. Draw circuits with a ruler, straight lines and square corners.

A simple circuit

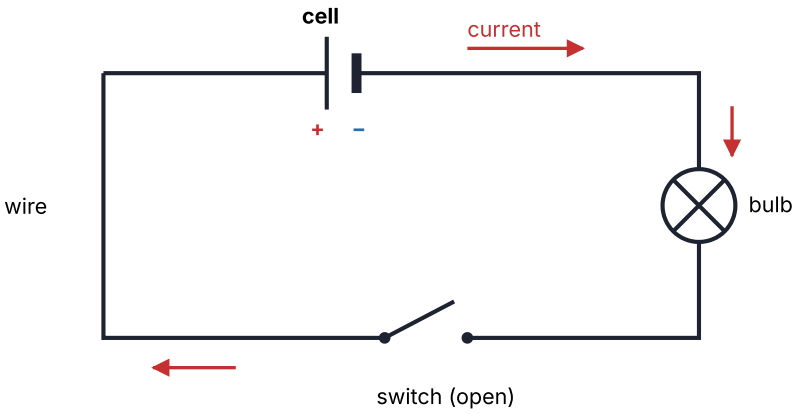


Figure 6: A cell, a switch and a bulb. The switch is open, so no current flows yet. When it closes, conventional current (red arrows) flows from + to – around the outside of the circuit.

The four key terms

Term	What it means (easy words)	Unit	Measured with
Charge (Q)	The "stuff" that flows in a wire – carried by electrons. Charge is **not used up**; it goes round and round. It is the **energy** that gets used.	coulomb (C)	–
Current (I)	How much charge flows past a point **every second** (rate of flow of charge).	ampere (A)	**Ammeter**, connected **in series**
Potential difference (V) (voltage)	The **energy given to each coulomb** of charge between two points. Think of it as the "push" of the cell.	volt (V)	**Voltmeter**, connected **in parallel** (across a component)
Resistance (R)	How much a component **opposes (slows down) the current**. More resistance = less current.	ohm (Ω)	Calculate: R = V ÷ I

Water pipe picture: the cell is like a pump, the **voltage** is how hard the pump pushes, the **current** is how much water flows per second, and **resistance** is a narrow part of the pipe that makes it harder for water to flow.

Example (Q = I × t): 120 C passes a point in 1 minute. Find the current.
t = 60 s → I = Q ÷ t = 120 ÷ 60 = **2 A**

Example (V = I × R): A voltmeter across a bulb reads 9 V and the ammeter reads 1.5 A. Find the resistance.
R = V ÷ I = 9 ÷ 1.5 = **6 Ω**

Example (V = I × R): A 12 V battery is connected to a 4 Ω resistor. Find the current.
I = V ÷ R = 12 ÷ 4 = **3 A**

4. Series and parallel circuits

In a **series** circuit there is only **one path** for the current. In a **parallel** circuit there are **two or more paths (branches)**.

Series circuit

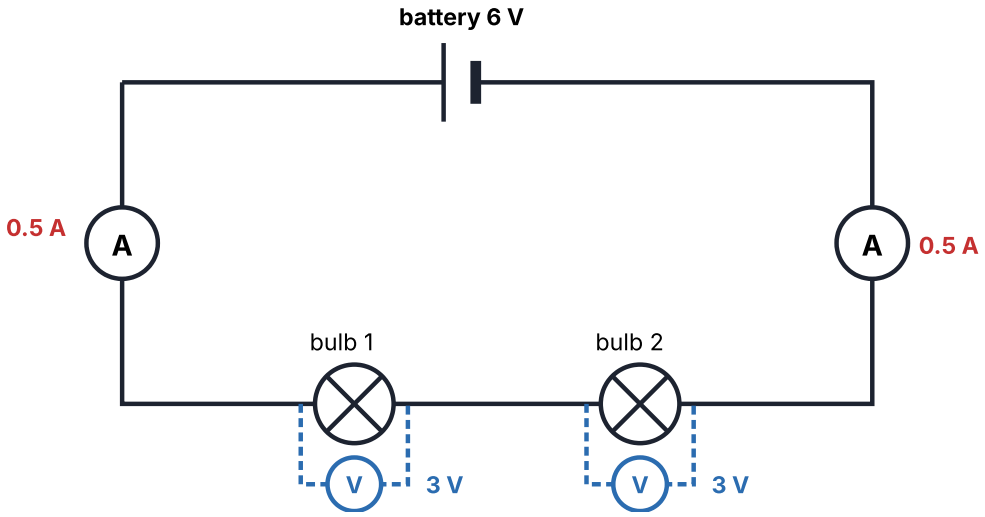


Figure 7: A series circuit. Both ammeters read the same (0.5 A). The 6 V is shared between the bulbs: 3 V + 3 V = 6 V.

Rule for series circuits	In symbols
The **current is the same everywhere** (only one path, so all the charge goes through everything)	$I = I_1 = I_2$
The **voltage is shared** between the components; the parts add up to the battery's voltage	$V = V_1 + V_2$
The **total resistance adds up**	$R = R_1 + R_2$
Add more bulbs → more resistance → less current → bulbs get **dimmer**	
One bulb breaks → the circuit is open → **all** the bulbs go out	

Series example: A 12 V battery with a 2 Ω and a 4 Ω resistor in series.
(a) Total resistance: R = 2 + 4 = **6 Ω**
(b) Current: I = V ÷ R = 12 ÷ 6 = **2 A** (same through both resistors)
(c) Voltage across 2 Ω: V = I × R = 2 × 2 = **4 V**
(d) Voltage across 4 Ω: V = 2 × 4 = **8 V** Check: 4 + 8 = 12 V ✓

Parallel circuit

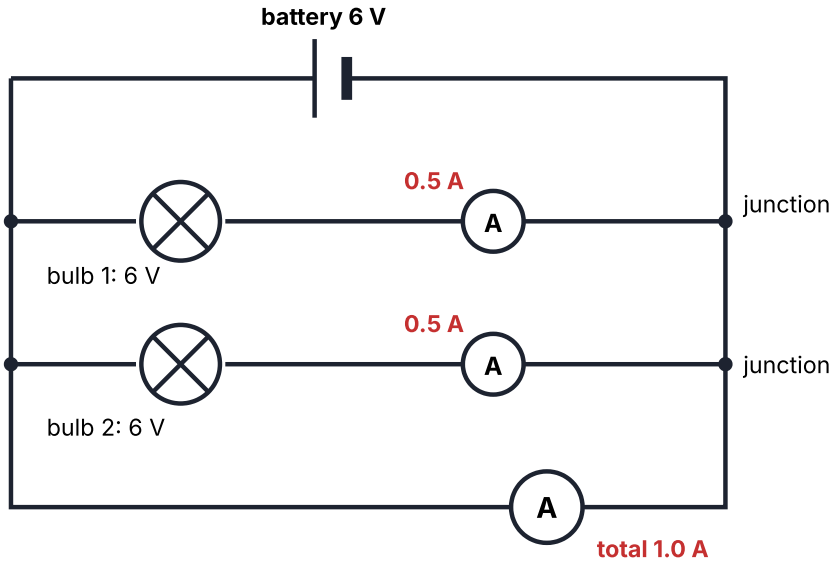


Figure 8: A parallel circuit with two branches. Each bulb gets the full 6 V. The current splits: 0.5 A + 0.5 A = 1.0 A from the battery.

Rule for parallel circuits	In symbols
The **voltage is the same** across every branch (each branch is connected straight to the battery)	$V = V_1 = V_2$
The **current splits** at a junction; the branch currents add up to the total	$I = I_1 + I_2$
More current flows through the branch with the **smaller resistance**	
Adding more resistors in parallel makes the **total resistance smaller** (more paths for the current)	$1/R = 1/R_1 + 1/R_2$
Add more bulbs → each stays **equally bright** (but the battery goes flat faster)	
One bulb breaks → the **others stay on**. That is why houses are wired in parallel.	

Parallel example: A 6 V battery with a 3 Ω and a 6 Ω resistor in parallel.
(a) Voltage across each resistor = **6 V**
(b) Current through 3 Ω: $I = V \div R = 6 \div 3 =$ **2 A**
(c) Current through 6 Ω: $I = 6 \div 6 =$ **1 A**
(d) Total current: 2 + 1 = **3 A**
(e) Total resistance: $R = V \div I = 6 \div 3 =$ **2 Ω** (smaller than 3 Ω ✓)

Example: The main ammeter reads 1.2 A and the ammeter in branch 1 reads 0.5 A. Find the branch 2 current.
$I = I_1 + I_2 \rightarrow 1.2 = 0.5 + I_2 \rightarrow I_2 =$ **0.7 A**

Series vs parallel at a glance

	Series	Parallel
Paths	One	More than one
Current	Same everywhere	Splits between branches
Voltage	Shared (adds up)	Same across each branch
Total resistance	Gets bigger as you add resistors	Gets smaller as you add resistors
One bulb blows	All go out	Others stay on
Example	Old Christmas lights	House wiring

5. What affects the resistance of a wire?

Resistance happens because moving electrons **bump into the atoms** of the wire. Anything that causes **more bumps** gives **more resistance**.

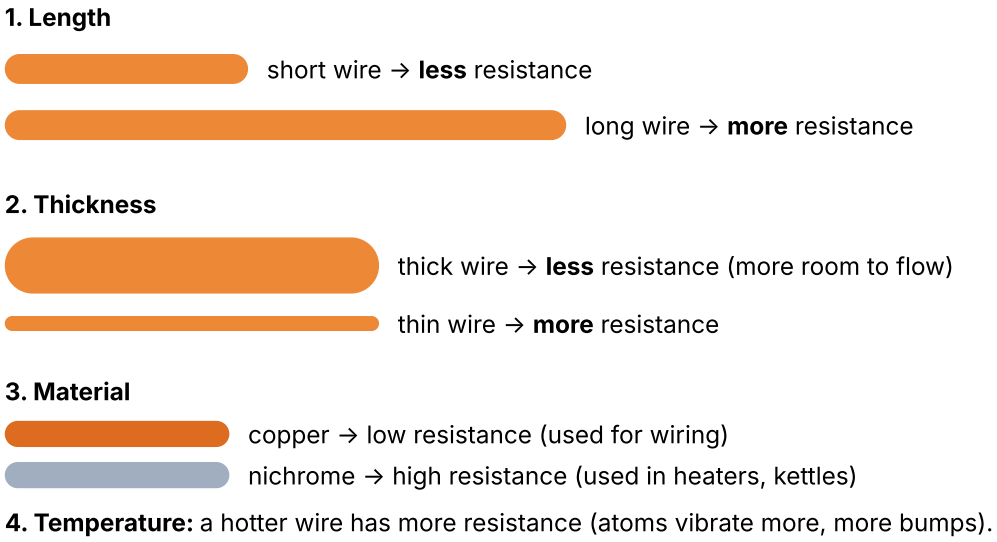


Figure 9: The four factors that affect resistance

- **Conductors** let current flow easily: metals (copper, aluminium, silver, iron), graphite, salt water.
- **Insulators** do not let current flow: plastic, rubber, wood, glass, dry air.
- Electric wires are **copper** (good conductor) covered in **plastic** (insulator, for safety).

> **Practical investigation:** to test how length affects resistance, change only the **length** (independent variable), measure the **current** (dependent variable), and keep the thickness, material, temperature and battery the **same** (controlled variables). Result: longer wire → lower ammeter reading → higher resistance.

6. Forces

A **force** is a **push or a pull** on an object. A force always has a **size** and a **direction**, so we draw it as an **arrow**. Force is measured in **newtons (N)** with a spring balance. (1 N is about the weight of a small apple.)

The 4 effects of forces

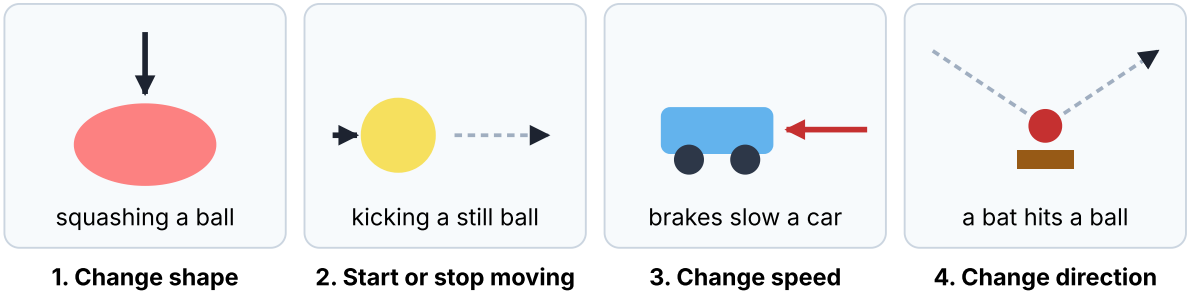
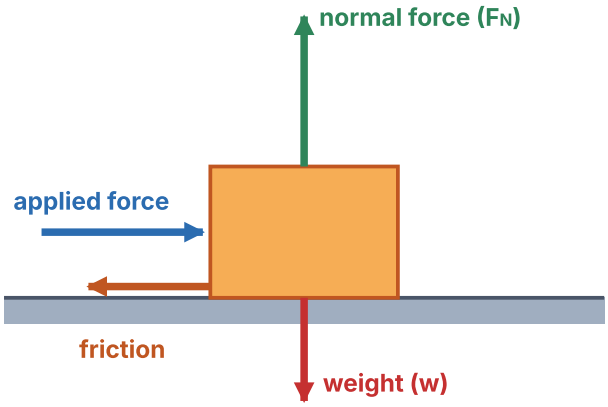


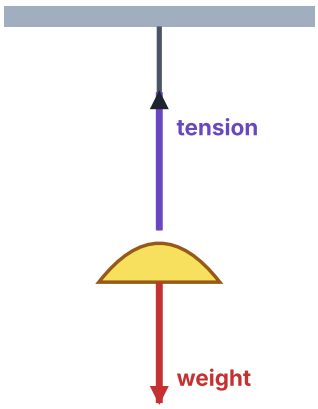
Figure 10: A force can change the shape, motion, speed or direction of an object

Contact forces and field forces

Contact forces only act when objects **touch**. **Field forces** (non-contact forces) act **from a distance**, without touching.



A box pushed to the right: friction acts to the left, against the motion



A hanging lamp: tension pulls up along the cable

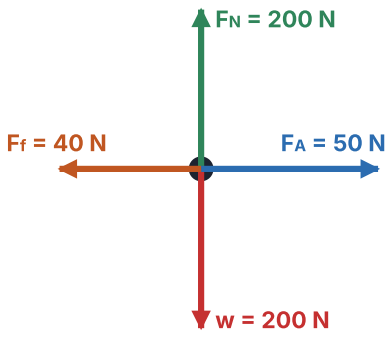
Contact force	Explanation	Direction
Applied force	A push or pull by a person or object (pushing a trolley)	The way you push or pull
Friction	Force between two surfaces rubbing together that **opposes motion**. Rough surfaces → more friction. It produces heat.	**Opposite** to the motion, along the surface
Normal force	The push of a surface on an object resting on it (a table pushing up on a book)	**Perpendicular (90°)** to the surface
Tension	The pull in a stretched rope, string or cable	Along the rope, away from the object

Field force	Explanation
Weight (gravitational force)	The Earth pulls every object towards its centre – always drawn **straight down**
Electrostatic force	Attraction or repulsion between **charged** objects (section 8)
Magnetic force	Attraction or repulsion between **magnets**, or a magnet and iron (section 7)

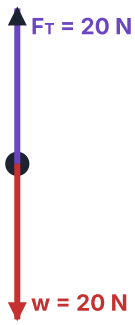
> **Friction is useful:** it lets us walk, gives tyres grip and makes brakes work. **Friction is a nuisance:** it wears out shoes and machine parts and wastes energy as heat. Reduce it with oil, grease, ball bearings and smooth or streamlined shapes.

Free body diagrams

A **free body diagram** shows the object as a **dot**, with an arrow for every force acting **on** it, drawn **from the dot**.



A box pushed right and speeding up: the applied force arrow is longer than friction



The hanging lamp: equal arrows, because the forces balance

Rules for force diagrams: (1) only forces acting **on** the object, (2) arrows point in the direction of the force, (3) **longer arrow = bigger force**, equal forces get equal arrows, (4) label every arrow (w, F_N, F_A, F_f, F_T), (5) use a ruler.

Mass and weight

Mass is the **amount of matter** in an object, measured in **kilograms (kg)**. **Weight** is the **force with which the Earth pulls** on the object, measured in **newtons (N)**. Your mass is the same everywhere, but your weight depends on how strong gravity is.

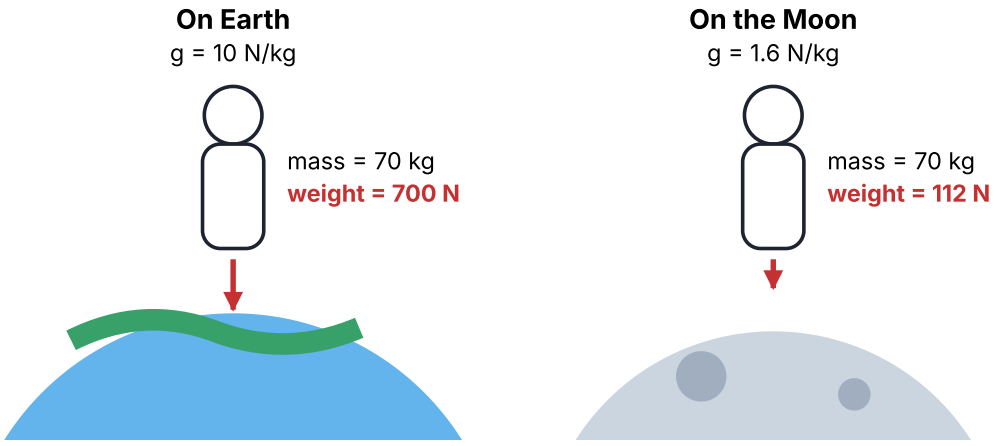


Figure 11: An astronaut has the same mass everywhere, but weighs about ⅙ as much on the Moon because the Moon's gravity is weaker

	Mass	Weight
Meaning	Amount of matter	Gravitational pull of the Earth on an object
Unit	kilogram (kg)	newton (N)
Measured with	Balance / scale	Spring balance
Is it a force?	No	Yes – a field force, acting downwards
Changes on the Moon?	No	Yes – about ⅙

w = m × g (g = 10 N/kg on Earth)
Weight of a 45 kg learner: w = 45 × 10 = **450 N**
Weight of a 250 g packet: m = 0.25 kg → w = 0.25 × 10 = **2.5 N**
Mass of a box that weighs 800 N: m = w ÷ g = 800 ÷ 10 = **80 kg**

In everyday life people say "I weigh 50 kg". In science, 50 kg is your **mass**; your weight is 500 N.

7. Magnetism

A **magnet** attracts magnetic materials such as iron. The **magnetic force** is a **field force** – it works without touching, and it gets weaker the further away you go.

Poles of a magnet

- Every magnet has a **north pole (N)** and a **south pole (S)**. The force is **strongest at the poles**.
- **Like poles repel** (N–N, S–S) and **unlike poles attract** (N–S).
- Poles always come in pairs. Cut a magnet in half and you get **two** smaller magnets, each with an N and an S pole.
- A hanging magnet (or a compass needle) turns until its N pole points **north**.
- **Repulsion is the only sure test for a magnet** – a magnet attracts iron *and* other magnets, but it can only repel another magnet.

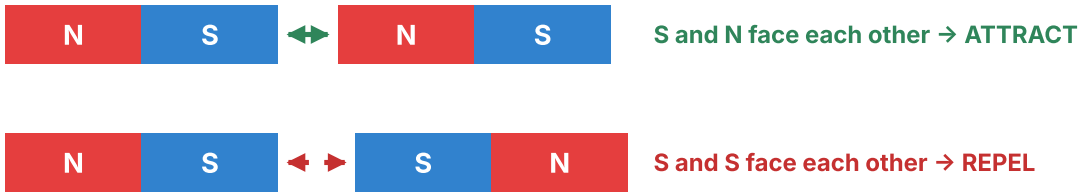


Figure 12: Unlike poles attract, like poles repel

Magnetic fields

A **magnetic field** is the area around a magnet where a magnetic force can be felt. You can see its shape by sprinkling **iron filings** around a magnet, or by moving a small **plotting compass** around it.

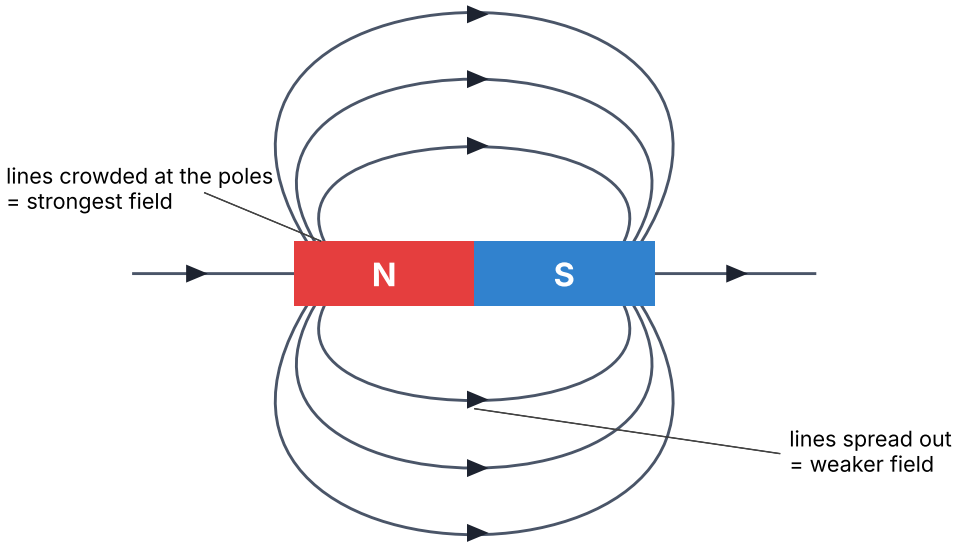
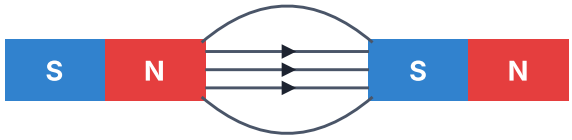
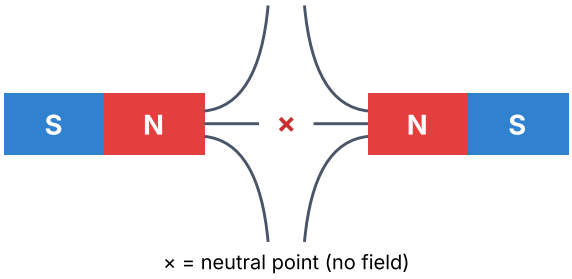


Figure 13: Magnetic field of a bar magnet. The arrows go from N to S outside the magnet.

Rules for drawing field lines: (1) arrows point **from N to S** outside the magnet, (2) lines **never cross**, (3) lines are **closest together at the poles**, where the field is strongest.



Unlike poles (N facing S): lines go straight across – attraction

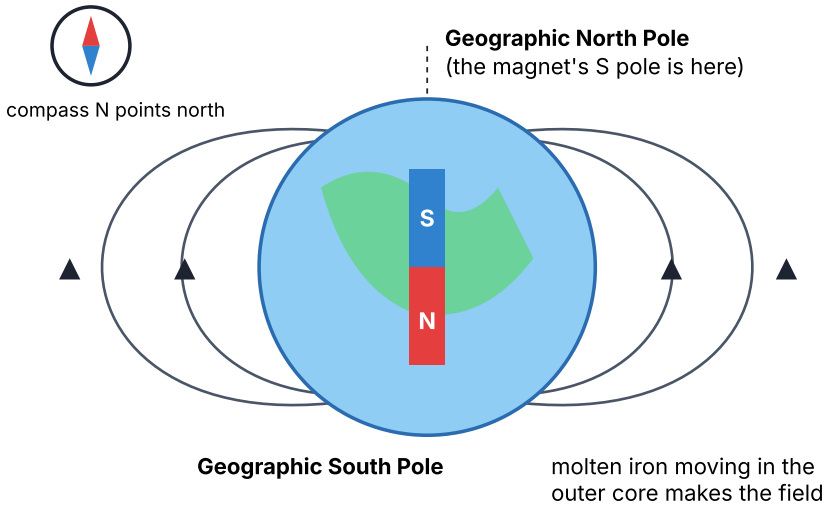


Like poles (N facing N): lines bend away – repulsion

Magnetic and ferromagnetic materials

- **Magnetic materials** are attracted to a magnet. **Ferromagnetic materials** are **strongly** attracted and can be made into magnets: **iron, nickel, cobalt** and their alloys such as **steel**.
- **Non-magnetic materials:** wood, plastic, glass, paper, rubber, and most metals – **copper, aluminium, gold, silver, brass**.
- ⚠ **Not all metals are magnetic.** An aluminium cold-drink can is not attracted; a steel food can is.
- **Soft iron** magnetises easily but loses its magnetism quickly (temporary magnet – used in electromagnets). **Steel** keeps its magnetism (permanent magnet – fridge magnets, compass needles).
- **To make a magnet:** stroke a steel nail many times in **one direction** with one pole of a magnet. **To destroy it:** heat it, hammer it or drop it.

The Earth's magnetic field



Field lines point up: out of the magnet's N pole (south) and into its S pole (north)

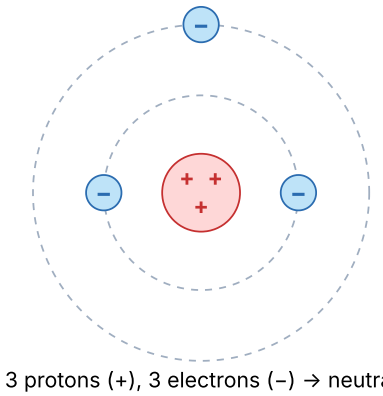
Figure 14: The Earth acts like a giant bar magnet. Its magnetic south pole lies near the geographic North Pole – so the N end of a compass is pulled north (unlike poles attract).

- The field is caused by **moving molten iron and nickel in the Earth's outer core**.
- The magnetic poles are not exactly at the geographic poles, and they slowly move.
- The magnetic field (the **magnetosphere**) **protects us from the solar wind** – harmful charged particles from the Sun.
- Near the poles some particles get in and make the sky glow: the **aurora** (northern and southern lights).
- Uses: navigating with a **compass**; birds and turtles use the field to find their way when they migrate.

8. Electrostatics

Electrostatics is about **static electricity** – charges that build up on objects and stay there. Objects become charged when **electrons move** from one object to another.

Charges in an atom



A neutral atom has equal + and − charges

Particle	Charge	Where	Can it move to another object?
Proton	+ positive	Nucleus (centre)	**No** – stuck in the nucleus
Neutron	none	Nucleus	No
Electron	− negative	Around the nucleus	**Yes** – the outer ones can be rubbed off

A **neutral** object has the same number of protons and electrons. Charge is measured in **coulomb (C)**.

How objects become charged: rubbing (friction)

When two different materials are **rubbed together**, **electrons move from one to the other**.

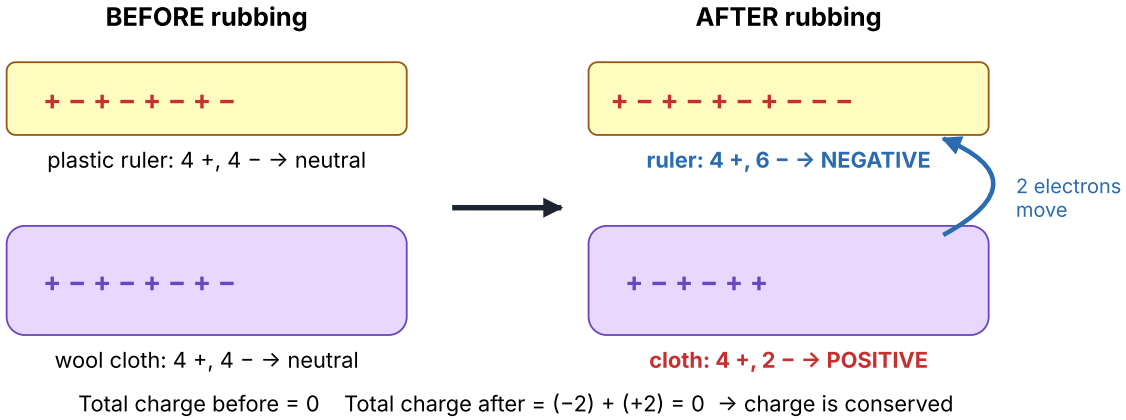


Figure 15: Rubbing a plastic ruler with a woollen cloth. Electrons move from the cloth to the ruler.

The object that…	ends up with…	so its charge is
gains electrons	more electrons than protons	**negative (−)**
loses electrons	fewer electrons than protons	**positive (+)**

Only electrons move. An object becomes positive by **losing electrons** – never by gaining protons. The two objects always end up with **opposite charges of the same size**.

Rubbed together	Becomes negative (gains e^-)	Becomes positive (loses e^-)
Plastic ruler + wool	Plastic ruler	Wool
Glass rod + silk	Silk	Glass rod
Balloon + hair	Balloon	Hair

The electrostatic force

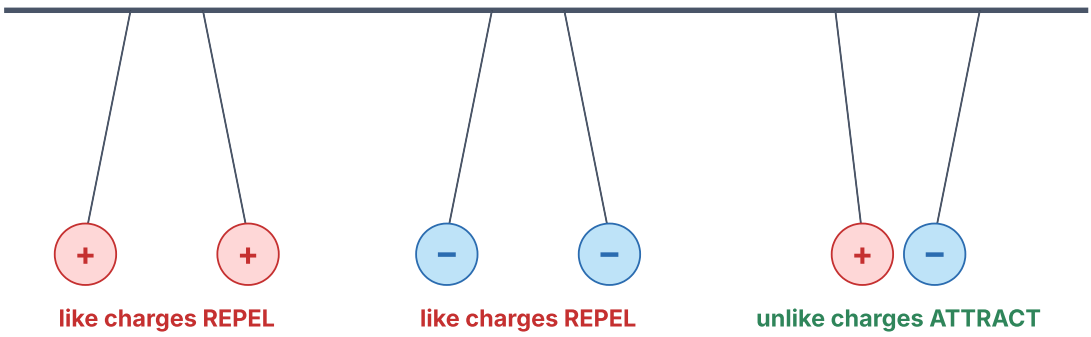


Figure 16: Charged balls hanging on threads

- The **electrostatic force** is the attraction or repulsion between charged objects. It is a **field force**.
- **Like charges repel**; **unlike charges attract**.
- The force is **stronger** when the objects are **closer** and when they carry **more charge**.
- A charged object also **attracts neutral objects**: a charged comb picks up small bits of paper, and a rubbed balloon sticks to a wall.

Conservation of charge

> **Charge cannot be created or destroyed – it can only be moved (transferred) from one object to another.** The total charge before and after rubbing is the same. In Figure 15, the ruler gained 2 electrons and the cloth lost exactly 2 electrons, so the total stays 0.

Static electricity in everyday life

- A small shock from a car door or after walking on a carpet; hair standing up after taking off a jersey.
- **Lightning** – charge builds up in thunderclouds and jumps to the ground as a giant spark.
- **Uses:** photocopiers and printers, spray-painting cars, dust filters in factory chimneys.
- **Dangers:** sparks can light petrol or gas – fuel trucks are **earthed** with a chain or cable. **Lightning conductors** on tall buildings carry charge safely into the ground.